A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Fluctuations in supply/demand & energy storage - Set 1

Mark scheme

AQA spec 3.3.4.1; Energy resources; textbook pp. 217-224

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Lowest overnight (about 25 GW at 04:00), rising steeply from 06:00 to about 40 GW by 08:00 **1** Fairly steady through the day then a peak of about 50 GW at 18:00 (evening) before falling overnight **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** 24-hour day / night cycle: people asleep overnight – morning rise as heating, lighting, cooking, industry and offices start **up to 2 Evening peak:** lighting, cooking, heating as people return home – short-term 'TV pickup' at programme breaks / mealtimes **up to 2 Other causes:** seasonal (winter heating / lighting, summer air conditioning) – weather-related (cold snaps) – weekday / weekend differences in industrial use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** They generate 24 hours a day because it is uneconomic / slow to turn them off and on, and output changes slowly **1** Demand falls faster at night than output can be reduced, so the surplus must be stored or is lost as heat from cables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
01.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Intermittent renewable resources:** wind, solar, tidal **1** Bulk delivery of transported fuels (oil tankers, coal trains, biofuel harvests) **1** Breakdowns / maintenance of power stations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 221-222; AQA spec 3.3.4.1.	

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Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** At times of low demand / surplus supply (night, windy periods) surplus electricity pumps water from the lower to the upper reservoir, storing it as gravitational potential energy **1** At peaks in demand water flows down through the turbines generating electricity **1** Can go from standby to full power in under 15 seconds so it meets sudden peaks (TV pickup) that fossil / nuclear stations cannot follow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Mass = $7 \times 10^6 \times 1000 = 7 \times 10^9$ kg **1** PE = m g h = $7 \times 10^9 \times 9.8 \times 500$ **1** = 3.43×10^{13} J (accept 3.4×10^{13}) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (J) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** The electricity used for pumping is surplus that would otherwise be wasted / lost as heat, or cheap off-peak electricity **1** It avoids building rapid-response low-efficiency peaking plants (open-cycle gas turbines) and allows intermittent renewables to be used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2

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Question	Marking guidance	Marks
02.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Two water bodies / sites with a large height difference close together (mountainous) **1** Impermeable, stable rock **1** Close to the grid / demand **1** Adequate water supply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222; AQA spec 3.3.4.1.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored **1** Hydrogen has a high energy density so it can be transported / stored for long periods **1** Released by combustion for heat, industrial processes or steam turbines **OR** by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water **1** Can also be used as a vehicle fuel or converted to methane / ammonia **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
03.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks **1** Liquefied at very low temperature - energy for refrigeration and compression **1** Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy **1** Conversion to ammonia - some energy needed but less than the alternatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging **1** A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **fuel cell** → a cell that produces electricity from hydrogen and oxygen	2
03.4	1 mark for the point below. **1** Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.	
04.1	**1 mark each** 1 mark for each of the 4 points below. **1 Vehicles:** high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater – many cycles and high efficiency **1 Stationary storage:** mass is unimportant so the low energy density of lead-acid does not matter – lowest cost per kWh – well-established, recyclable **1 Lead-acid drawbacks:** fewer cycles and lower efficiency – lithium fire risk needs management **1** Nickel-cadmium largely replaced because of cadmium toxicity and cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

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Question	Marking guidance	Marks
04.2	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Recharging speed **1** Self-discharge during non-use **1** Availability of raw materials (lithium, cobalt) **1** Recyclability / disposal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Parked electric vehicles plugged into the grid form a very large distributed battery **1** At peaks a small proportion of their stored energy is fed back to the grid – surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surplus electricity electrolyses water to hydrogen, which can be converted to methane **1** Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Surplus energy drives a pump compressing air stored in underground caverns (salt mines) – releasing it drives machinery / turbines **1** Compression produces a lot of heat – storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** As volume increases the surface area : volume ratio decreases **1** Heat is lost through the surface so there is less surface per unit of stored heat – very large stores can carry summer solar heat into winter (inter-seasonal storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation **1** Water could only reach such temperatures under high pressure, increasing costs – high temperature needed to raise steam for turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Materials with high specific heat capacity (concrete, water) warm and cool slowly **1** They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Flywheels:** rotating mass stores kinetic energy for short-term smoothing / generation **1 Supercapacitors:** electrochemical capacitors with 10-100 times the energy density of conventional capacitors – possible future large-scale electricity storage **1** Thermal storage in molten salt / high-volume tanks – hydrogen / P2G – batteries **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.	

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Question	Marking guidance	Marks
06.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable – demand fluctuates daily, weekly, seasonally and short-term – supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants – large-scale electricity storage not yet possible directly so energy must be converted • **Pumped-storage HEP:** proven, rapid response, peak shaving, but limited sites • **Batteries:** lithium-ion for grid and vehicles – factors affecting viability – V2G • **Hydrogen:** electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid – high energy density enables vehicle use • Compressed air with heat recovery – thermal storage (molten salt for CSP, high-volume stores, thermal mass) – flywheels – supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage – energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

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Question	Marking guidance	Marks
	• **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **HEP** → electricity generated from falling water • **energy conservation** → using less energy, eg by insulation or behaviour change **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (9 marks)	
	Renewable electricity is intermittent - solar stops at night, wind on calm days, tidal power at slack water - and some of it is unpredictable, while demand fluctuates by the minute, day, week and season, so supply and demand rarely coincide. Without storage, surplus renewable output is wasted and shortfalls must be met by fossil-fuel peaking plants, which defeats the purpose. Electricity itself cannot be stored at scale, so the energy must be converted into another form and back.	
	Pumped-storage HEP is proven and responds within seconds for peak shaving, but suitable sites are few. Lithium-ion batteries now work at grid scale and in vehicles, with vehicle-to-grid schemes turning car batteries into distributed storage, though lithium supplies and lifetime limit them. Hydrogen from electrolysis can be stored, used in fuel cells or injected into the gas grid (power-to-gas), and its high energy density suits vehicles, although each conversion loses energy. Compressed air with heat recovery, molten-salt thermal stores for concentrating solar plants, thermal mass in buildings, flywheels and supercapacitors each fit particular timescales.	
	Storage is not the only answer: a diverse mix of wind, solar, tidal and biofuel smooths supply, smart-grid management shifts demand, and conservation reduces peaks. But these reduce rather than remove the mismatch. Short-term storage can be batteries and pumped hydro; the harder problem is seasonal storage, where only hydrogen or fuels made from surplus electricity look plausible. Without both, renewables can supplement but not dominate a country's supply.	
	Weaker answer - Level 2 (5 marks)	
	Renewable energy is intermittent. Solar panels do not work at night and wind turbines do not work when it is not windy, but people still need electricity then. Storage means the energy can be saved for when it is needed. Pumped storage pumps water uphill when there is spare electricity and lets it flow back through turbines at peak times. Batteries can also store electricity. Hydrogen can be made from electricity and used in fuel cells. So storage is needed for renewables to provide most electricity.	
	What makes the difference	
	• Level 2 explains the mismatch and describes three storage methods correctly, but does not evaluate them (sites, lithium, conversion losses, timescales).	
	• Level 3 distinguishes short-term from seasonal storage - the key insight for 'most of a country's electricity'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 also considers the alternatives to storage (diversity, grid management, conservation) and explains why they are insufficient alone.	
06.2	1 mark for the point below. **1** Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

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